Frank Schlehofer

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Education

Purdue University West Lafayette, IN

Bachelor of Science, Master of Science in Computer Science *December 2027*

- GPA: 3.99/4.00
- Relevant Coursework: Data Communication and Networks, Compilers, Analysis of Algorithms, Data Mining and Machine Learning, Artificial Intelligence, Data Structures and Algorithms, Systems Programming, Computer Architecture, Relational Databases, Object-Oriented Programming, Discrete Math, C Programming, Statistics, Probability, Linear Algebra

Experience

Software Engineering Intern May 2026 – Aug 2026

Garmin International | Python, Unit/Integration Testing, CI/CD, Git, Jira, Github Copilot

- Built features for internal automation tooling using Python to accelerate complex data transformations, streamline development workflows, and enhance overall engineering efficiency
- Optimized core internal automation workflows by designing and implementing structural enhancements that reduced data lookup and transformation times by up to 50%, accelerating key engineering functions
- Developed comprehensive unit, integration, and regression tests, guaranteeing the high-fidelity validation of critical aircraft performance datasets prior to deployment

Undergraduate Teaching Assistant Jan 2025 – Present

Purdue University | C, Java, SQL, MySQL, MongoDB, Relational Databases

- Mentored and instructed a cohort of 700+ students on foundational C, Java, and relational database concepts through 3+ weekly office hours and lab sessions
- Graded and provided individualized feedback on over 2,500 programming assignments, identifying common errors to help improve departmental teaching materials
- Directly assisted an average of 30+ students per week with complex topics including memory management in C, polymorphism in Java, and algorithmic debugging

Team Lead, Flight Simulation Sub-Team – SlayterHiL Aug 2025 - Present

Purdue Embedded Systems Club | C, C++, Protobuf, SPI, Git, Makefile, Raspberry Pi

- Led an 8-person flight simulation sub-team within a hardware-in-the-loop (HIL) testing platform for a custom quadcopter flight controller
- Engineered the team's data communication pipeline, serializing simulation data with Protobuf and streaming it over SPI to onboard hardware, enabling real-time data exchange between the simulation and physical flight controller

Projects

Frankie's Fridge | *Node, React, Express, REST API, Agile, JavaScript, Google Cloud Vision API, OpenAI API*

- Developed full-stack web application, using Node.js and Express for backend server with RESTful API endpoints, PostgreSQL for persistent and scalable data storage, and React.js styled with CSS for intuitive fridge UI
- Drove user adoption to 30+ active registrations within the first month post-launch through LinkedIn marketing
- Integrated Google Cloud Vision API to enable image-based ingredient recognition, reducing manual user input by an estimated 75% and enhancing the core user experience

Hybrid Movie Recommendation System | *Python, PyTorch, Deep Learning, Machine Learning, Pandas, NumPy*

- Built a deep learning-based collaborative filtering recommendation system from scratch in PyTorch, implementing Matrix Factorization with embedding layers and engineering the full ML pipeline (data preprocessing, custom Dataset/DataLoader, training/validation loop)
- Diagnosed and resolved key model performance issues including the cold-start problem, unconstrained outputs, and overfitting via activation functions and weight decay

Technical Skills

Programming: Python, C, Java, C++, JavaScript, Typescript, SQL, HTML, CSS, x86 Assembly, Yacc, Lex, Swift
Frameworks/Tools: React, Node, Express, PostgreSQL, MySQL, MongoDB, Git, GitHub, Agile, OpenCV, TensorFlow, PyTorch, Numpy, Pandas, Scikit-learn, LLVM, Django, Github Copilot, ChatGPT
Technologies: AWS, Google Cloud Vision API, LaTeX, Supabase, Render, Vercel, Netlify